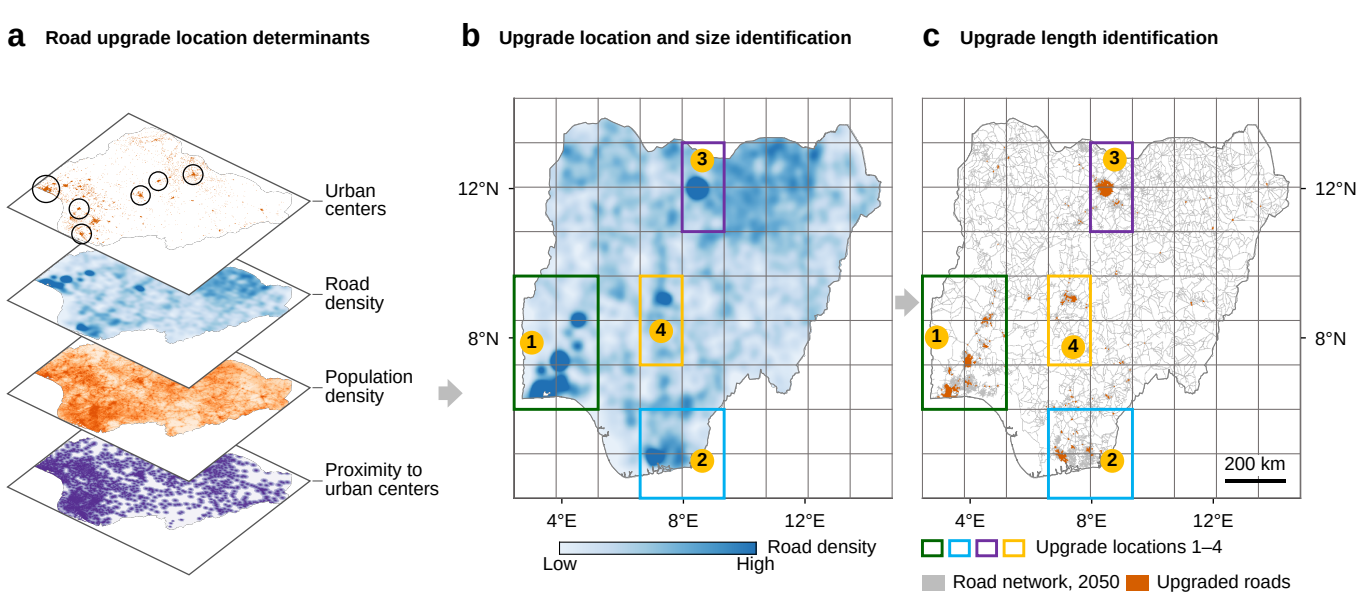

a
Road upgrade location determinants
Urban centers
Road density
Population density
Proximity to urban centers
b
Upgrade location and size identification
12°N
8°N
4°E
8°E
12°E
1
2
3
4
Road density
Low
High
c
Upgrade length identification
12°N
8°N
4°E
8°E
12°E
200 km
Upgrade locations 1–4
Road network, 2050
Upgraded roads